CIE AS PHYSICS (9702) • QUESTION PAPER

3.1 Momentum & Newton's Laws of Motion

Syllabus Section: 3. Dynamics **Paper Type:** Multiple Choice (Paper 1) **Total:** 23 Questions (23 Marks)

Question 1 Easy [1 mark]

A tennis ball is dropped from different heights and its velocity is measured just before it hits the ground.

A graph is plotted showing the momentum of the tennis ball against its velocity.

What does the gradient of this graph represent?

A the acceleration of the tennis ball

B the mass of the tennis ball

C the kinetic energy of the tennis ball

D the velocity of the tennis ball

Question 2 Easy [1 mark]

A cyclist is riding at a steady speed on a level road

According to Newton's third law of motion, what is equal and opposite to the backward push of the back wheel on the road?

A the force exerted by the cyclist on the pedals

B the total air resistance and friction force

C the tension in the cycle chain

D the forward push of the road on the back wheel

Question 3 Easy [1 mark]

A resultant force causes a body to accelerate.

What is equal to the resultant force?

A the acceleration of the body per unit mass

B the change in kinetic energy of the body per unit time

C the change in momentum of the body per unit time

D the change in velocity of the body per unit time

Question 4 Easy [1 mark]

What describes the mass and the weight of an object?

	Mass	**Weight**
A	the ratio of the object's momentum to its velocity	the work done in lifting the object 1.0 m vertically
B	the force the object experiences due to gravity	the resistance of the object to changes in motion
C	the resistance of the object to changes in motion	the force the object experiences due to gravity
D	the total number of atoms in the object	the ratio of the object's mass to the acceleration of free fall

A

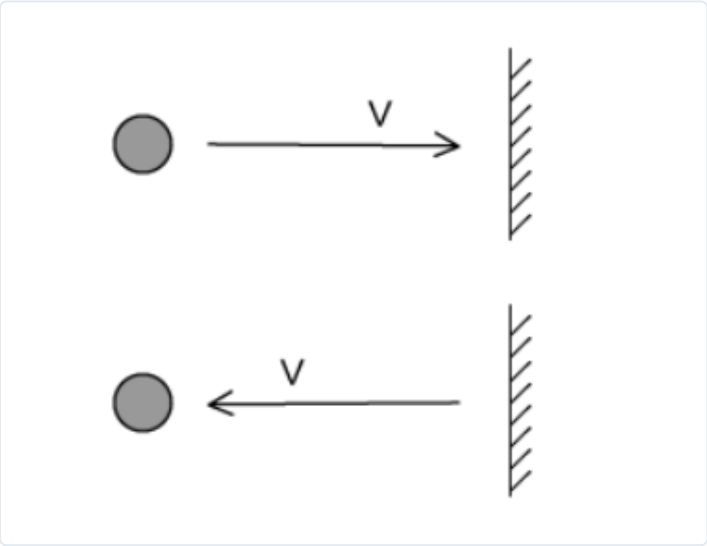
B

C

D

Question 5 Easy [1 mark]

An object travelling with velocity *v* strikes a wall and rebounds as shown



Which property of the object is **not** conserved?

A momentum

B mass

C kinetic energy

D speed

Question 6 Easy [1 mark]

A force *F* is applied to a freely moving object. At one instant of time, the object has velocity *v* and acceleration *a*.

Which quantities must be in the same direction?

A *a* and *v* only

B *v*, *a* and *F*

C *v* and *F* only

D *F* and *a* only

Question 7 Easy [1 mark]

Which is **not** one of Newton's laws of motion?

A if body A exerts a force on body B, then body B exerts an equal and oppositely-directed force on body A

B the rate of change of momentum of a body is directly proportional to the external force acting on the body and takes place in the direction of the force

C the total momentum of a system of interacting bodies remains constant, providing no external force acts

D a body continues in a state of rest or of uniform motion in a straight line unless acted upon by some external force

Question 8 Easy [1 mark]

The acceleration of free fall is represented by g.

Which statement is correct?

A g is the force due to gravity

B g is reduced by air resistance

C g is the weight of an object

D g is the ratio of an object's weight to its mass

Question 9 Easy [1 mark]

As a car accelerates uniformly, its momentum is measured at regular intervals, and a graph of the car's momentum is plotted against time.

Which quantity is equal to the gradient of the graph at a particular time?

A the acceleration of the car

B the resultant force on the car

C the kinetic energy of the car

D the velocity of the car

Question 10 Medium [1 mark]

All external forces on a body cancel out.

Which statement **must** be correct?

A the body does not move

B the total energy (kinetic and potential) of the body remains unchanged

C the speed of the body is zero

D the momentum of the body remains unchanged

Question 11 Medium [1 mark]

A ball falls vertically and bounces on the ground.

Which statement about the forces acting while the ball is in contact with the ground is correct?

A the force that the ball exerts on the ground is always equal to the weight of the ball

B the force that the ball exerts on the ground is always less than the weight of the ball

C the force that the ball exerts on the ground is always equal in magnitude and opposite in direction to the force the ground exerts on the ball

D the weight of the ball is always equal in magnitude and opposite in direction to the force that the ground exerts on the ball

Question 12 Medium [1 mark]

The acceleration of free fall on planet X is $\frac{1}{6}$ of the acceleration of free fall on Earth.

The mass of a body on planet X is 60 kg.

What is its weight on planet X?

A 10 N

B 59 N

C 98 N

D 360 N

Question 13 Medium [1 mark]

Newton's third law of motion is often summarised as

'Every action has an equal and opposite reaction.'

A cup of tea rests on a table. If the weight of the cup is the 'action' force, what is the 'reaction' force?

A the push of the table on the cup

B the push of the cup on the table

C the pull of the Earth on the cup

D the pull of the cup on the Earth

Question 14 Medium [1 mark]

The gravitational field strength on the surface of planet R is one eighth of that on the surface of planet S.

On the surface of R, a body has its mass measured to be 2.5 kg and its weight measured to be 12.5 N

What results are obtained for measurements of the mass and weight of the same body on the surface of planet S?

	Mass on S	Weight on S
A	2.5 kg	20 N
B	20 kg	100 N
C	2.5 kg	100 N
D	20 kg	160 N

A

B

C

D

Question 15 Medium [1 mark]

Water is pumped through a hose-pipe at a rate of 80 kg per minute. It emerges from the hose-pipe horizontally with a speed of $30 \quad m \quad s^{-1}$

Which force is required from a person holding the hose-pipe to prevent it moving backwards?

A 3 N

B 40 N

C 160 N

D 2400 N

Question 16 Medium [1 mark]

A car is travelling at constant velocity on a straight road.

Which row gives the correct Newton's third law force pair?

	Force A	Force B
A.	Driving force from engine	Air resistance on the car
B.	Weight of the car	Normal reaction from the road
C.	Driving force from wheels on road	Friction force from air on car
D.	Weight of the car	Gravitational force of the car on the Earth

A

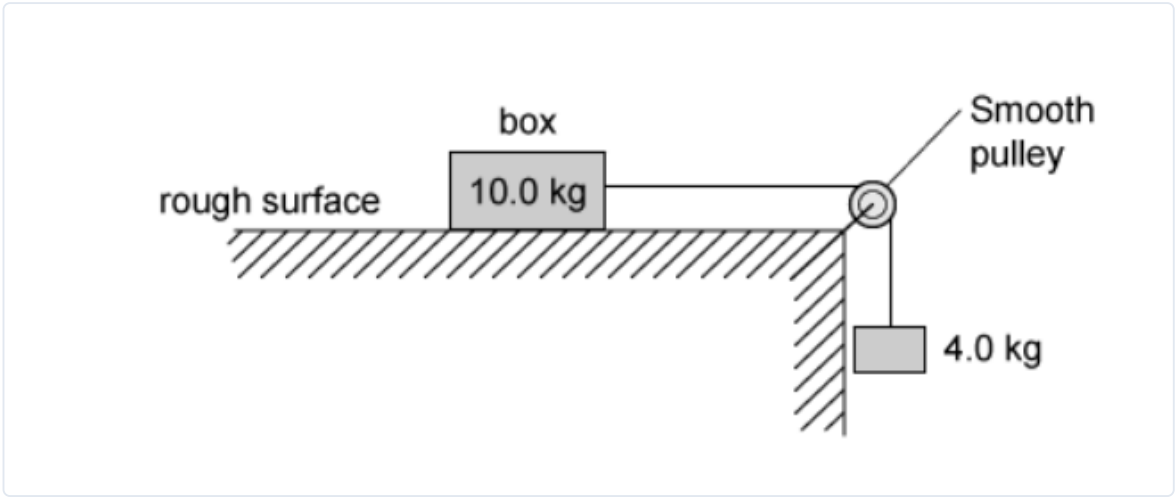
B

C

D

Question 17 Hard [1 mark]

A box of mass 10.0 kg rests on a horizontal rough surface. A string attached to the box passes over a smooth pulley and supports a 4.0 kg mass at its other end.



When the box is released, a frictional force of 12.0 N acts on it.

What is the acceleration of the box?

A 1.2 m s^{-2}

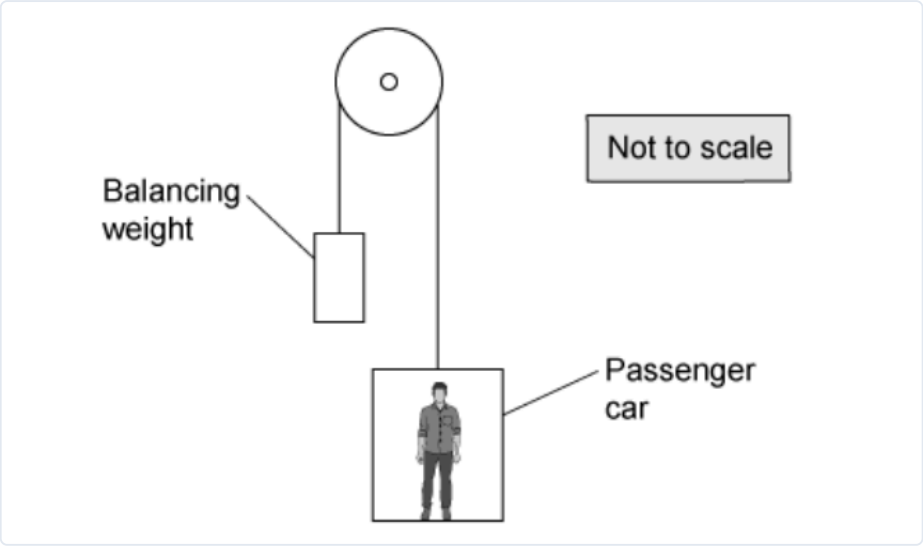
B 1.9 m s^{-2}

C 2.7 m s^{-2}

D 3.0 m s^{-2}

Question 18 Hard [1 mark]

A lift consists of a passenger car supported by a cable which runs over a light, frictionless pulley to a balancing weight. The balancing weight falls as the passenger car rises.



Some masses are shown in the table.

	Mass / kg
passenger car	1200
balancing weight	1320
passenger	80

What is the magnitude of the acceleration of the car when carrying just one passenger and when the pulley is free to rotate?

A $0.15\ m\ s^{-2}$

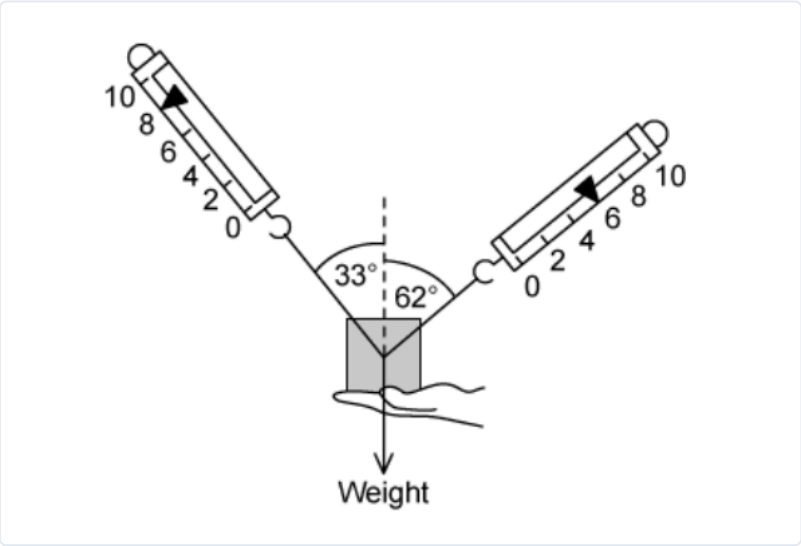
B $0.30\ m\ s^{-2}$

C $0.45\ m\ s^{-2}$

D $0.92\ m\ s^{-2}$

Question 19 Hard [1 mark]

A 3.2 kg mass is supported by a person's hand and two newton-meters as shown.



When the person's hand is removed, what is the initial vertical acceleration of the mass?

A $3.0\ \text{m s}^{-2}$

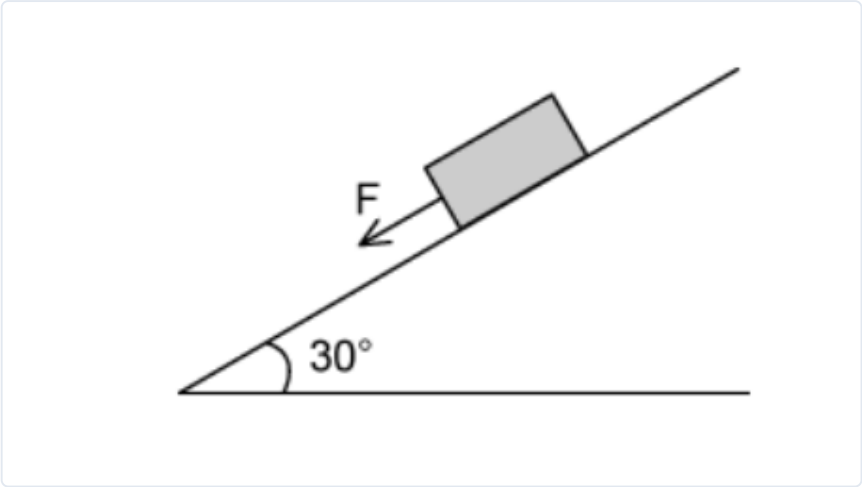
B $5.4\ \text{m s}^{-2}$

C $6.8\ \text{m s}^{-2}$

D $12.8\ \text{m s}^{-2}$

Question 20 Hard [1 mark]

A brick of mass 1.8 kg rests on a plane inclined at an angle of 30°. The weight of the brick has a component of *F* N parallel with the plane. The brick also experiences a frictional force of 3.0 N



What is the acceleration of the brick down the plane?

A $3.2\ m\ s^{-2}$

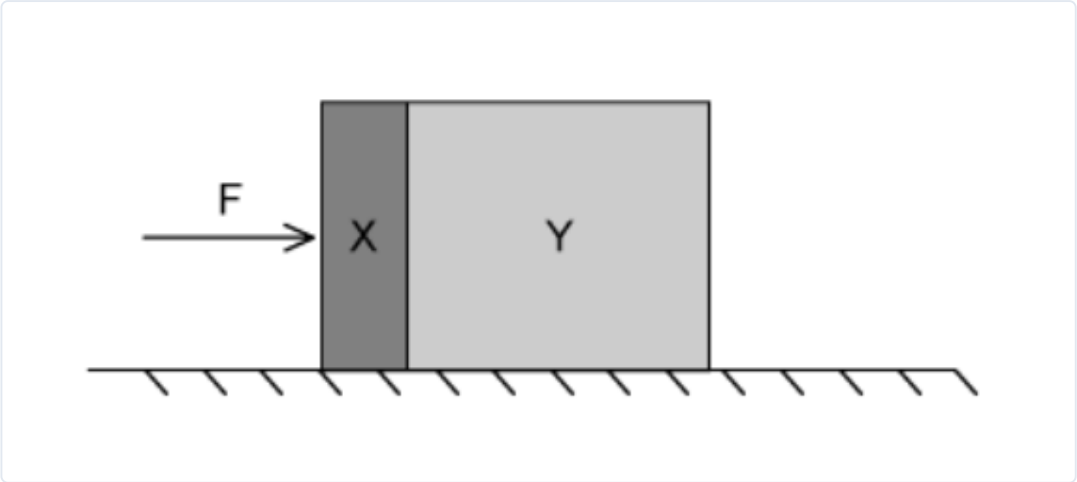
B $3.9\ m\ s^{-2}$

C $6.6\ m\ s^{-2}$

D $8.1\ m\ s^{-2}$

Question 21 Hard [1 mark]

Two blocks X and Y, of masses *m* and 4 *m* respectively, are accelerated along a smooth horizontal surface by a force *F* applied to block X as shown



What is the magnitude of the force exerted on block Y by block X during this acceleration?

A $\frac{F}{4}$

B $\frac{F}{5}$

C $\frac{4F}{5}$

D $\frac{5F}{4}$

Question 22 Hard [1 mark]

A ship of mass 6.7×10^7 kg is approaching a harbour with speed 18.9 m s^{-1}

By using reverse thrust it can maintain a constant total stopping force of 880,000 N

How long will it take to stop?

A 14 seconds

B 140 seconds

C 24 minutes

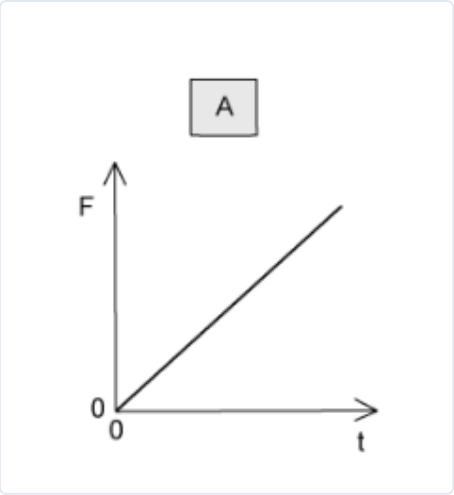
D 240 minutes

Question 23 Hard [1 mark]

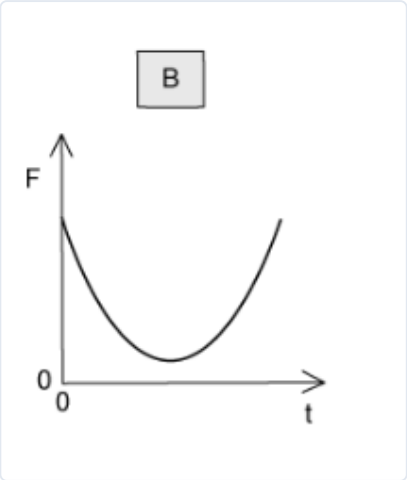
A tennis ball is dropped onto a table and bounces back up. The table exerts a force *F* on the ball.

Which graph best shows the variation with time *t* of the force *F* while the ball is in contact with the table?

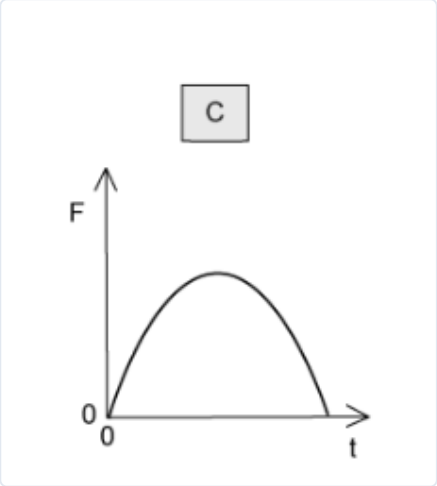
A



B



C



D

